



SQU Internship Written Report

By :

Mallak Al- Hakamani

Table of Contents :

1. Introduction.....	3
2. Background of Company.....	3
3. Summary of Duties.....	5
4. Working Experiences.....	5
5. Conclusion.....	5
6. References.....	6

● Introduction

According to the degree plan, in order to graduate, students must finish a training program. This training is crucial because it makes the connection between what we study in college and what we encounter in the workplace. I completed my training at the Sultan Qaboos University during the summer from June 15, 2025, to July 24, 2025, a period of six weeks.

During this period, I had the opportunity to training with alongside professionals in various IT domains, gaining exposure to networking, security, automation tools, and real-world IT infrastructure. The internship provided hands-on experience and enhanced my understanding of how academic concepts are applied in professional environments.

● Background of Company

Sultan Qaboos University (SQU) is the premier academic institution in Oman, established in 1986. Located in Al-Khoudh, Muscat, the university houses multiple colleges and research centers supported, by advanced technological infrastructure that facilitates both academic and administrative functions.

During the internship, I was assigned to the Information System, which is part of the university's IT services. This department is responsible for overseeing and supporting the university's internal systems, including servers, user accounts, security policies, software deployment, and data services. The internship offered hands-on exposure to a range of practical tasks, tools, and real-world challenges managed by the department.

● Summary of Duties

A) Week 1 :

During the first week, I attended a workshop on CV writing and job interviews. we were introduced to some basic information about cyber threats, vulnerability scanning tools, and practiced using Wireshark with VPNs for packet analysis. I also explored AI automation tools and Active Directory as initial concepts. In preparation for the second week, the plan with practical networking tasks, such as IP configuration and VLAN creation.

B) Week 2 :

In the second week, I trained on IP addressing and switch configuration, created VLANs, and physically connected switches using Ethernet cables. I also learned how to properly cut and organize console cables, and visited the university's VR lab and data center to explore real network environments and different cable types. Once challenge I faced was correctly cutting and ordering console cables, which required precision and practice.

C) Week 3 :

This week was focused on PowerShell scripting and Active Directory. I learned how to create, rename, and manage folders and files using PowerShell. I learned how to create AD users and groups and applied permissions and group policies. I also learned how to share folders to specific groups. The main challenge was understanding PowerShell syntax, but with repetition, it became manageable.

D) Week 4 :

This week focused on practicing Linux system management. We created a virtual machine with a Linux server installed. As a part of the training, we also learned how to create and manage hard disks in Linux. We continued practicing Linux commands and tested server access using tools like Putty. One challenge we faced was restricted access due to firewall rules, which limited our ability to connect to the server.

Additionally, I submitted an online certificate titled "**Cyber Security Attack Techniques**", which supported the hands-on experience by reinforcing key cyber security concepts.

Looking ahead to week 5, I planned to explore advanced networking tools, fiber cable types, and enterprise systems such as Aruba ClearPass.

E) Week 5 :

This week offered a variety of activities. I practiced AI automation using n8n tools to summarize news and designed a poster in the design workshop using Adobe Illustrator. I explored DAC cables, LC and fiber connectors and visited the data center to examine network setups. I also attended sessions about ClearPass, AirWave and Aruba Mobility Master and learned about core switch, patch panels and ISL cables. The variety of tasks this week made it both engaging and challenging.

F) Week 6:

The final week was a chance to review everything I had learned. I observed lab operations, supported basic troubleshooting and gained exposure to how the IT team manages internal documentation and maintenance. This week tied everything together and provided a solid conclusion to the internship. As a final step, I focused on summarizing my experiences and preparing to present what I learned in the final presentation.

- **Working Experiences**

Throughout the six-week internship at Sultan Qaboos University ,I gained valuable hands-on experience that enhanced both my technical and academic understanding .One of the most important experiences was working with networking devices such as switches , routers ,and cables ,and learning how to configure them in both simulated and real environments. I also developed a strong understanding of network security tools and how to analyze data packets using Wireshark.

Additionally, I improved my scripting skills using PowerShell, especially when creating and managing user accounts and permissions in an Active Directory environment. I had the opportunity to explore automation concepts using AI tools gained exposure to network monitoring and management platforms like Aruba ClearPass and AirWave.

Challenges included learning new tools and troubleshooting configuration issues, but through repetition and guided practice, I overcame them. These experiences helped bridge the gap between university coursework and actual fieldwork in the IT domain.

- **Conclusion**

The six-weeks internship at Sultan Qaboos University provided a valuable chance to immerse myself in the IT environment within a calm and academic setting. Since there were no fixed daily tasks, most of the learning happened through scheduled training sessions, individual practice and direct observation. I was able to follow how certain system were managed and understand the role of tools like Wireshark, Active Directory and PowerShell in real scenarios. This helped me connect what I studied at university with how it is actually used in the field.

The work environment was welcoming and supportive. Because the training took place inside a university, the overall atmosphere felt more educational than professional. We were treated more like students attending technical sessions than as employees, which made it easier to ask questions and absorb new information without pressure.

● References

1. Internal Training Materials - Sultan Qaboos University IT Department
2. Let's Defend Cybersecurity Lab - <https://letsdefend.io/>
3. n8n Automation Tool - <https://n8n.io/>
4. Adobe Illustrator - Used during design workshop
5. Ethernet Cable Preparation - Guided by staff and supported by online tutorials
6. Personal Notes - Collected during the six-week internship